

1. (c)

We know that,  $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$   
 $\left(1 + \frac{1}{x}\right)^n = {}^nC_0 + {}^nC_1\frac{1}{x} + {}^nC_2\frac{1}{x^2} + \dots + {}^nC_n\frac{1}{x^n}$  Obviously, the

term independent of  $x$  will be

$${}^nC_0 \cdot {}^nC_0 + {}^nC_1 \cdot {}^nC_1 + \dots + {}^nC_n \cdot {}^nC_n = C_0^2 + C_1^2 + \dots + C_n^2$$

**Trick :** Put  $n=1$  in the expansion of

$$(1+x)^1 \left(1 + \frac{1}{x}\right)^1 = 1+x + \frac{1}{x} + 1 = 2+x + \frac{1}{x} \dots (i)$$

We want coefficient of  $x^0$ . Comparing to equation (i).  
 Then, we get 2 i.e., independent of  $x$ .

Option (c) :  $C_0^2 + C_1^2 + \dots + C_n^2$ ; Put  $n=1$ ;

$$\text{Then } {}^1C_0^2 + {}^1C_1^2 = 1+1=2.$$

2. (b)

Coefficient of  $x^n$  in  $(1+x)(1-x)^n$  = Coefficient of  $x^n$  in  
 $(1-x)^n$  + coefficient of  $x^{n-1}$  in  $(1-x)^n$   
 = Coefficient of  $x^n$  in  $[{}^nC_n(-x)^n + x \cdot {}^nC_{n-1}(-x)^{n-1}]$   
 =  $(-1)^n {}^nC_n + (-1)^{n-1} \cdot {}^nC_1 = (-1)^n + (-1)^n \cdot (-n) = (-1)^n [1-n]$ .

3. (b)

$$\text{We have } [(x+3y)(3x-y)]^6 = [3x^2 + 8xy - 3y^2]^6;$$

$$\text{Number of terms} = \frac{(6+1)(6+2)}{2} = 28$$

4. (a)

$$\frac{1}{(1-x)(3-x)} = (1-x)^{-1}(3-x)^{-1} = 3^{-1}(1-x)^{-1} \left(1 - \frac{x}{3}\right)^{-1}$$

$$= \frac{1}{3} [1+x+x^2+\dots+x^n] \left[1 + \frac{x}{3} + \frac{x^2}{3^2} + \dots + \frac{x^{n-1}}{3^{n-1}} + \frac{x^n}{3^n}\right]$$

$$\text{Coefficient of } x^n = \frac{1}{3^{n+1}} + \frac{1}{3^n} + \frac{1}{3^{n-1}} + \dots + (n+1) \text{ terms}$$

$$= \frac{1}{3^{n+1}} \frac{[3^{n+1}-1]}{3-1} = \frac{3^{n+1}-1}{2 \cdot 3^{n+1}}$$

**Trick:** Put  $n=1, 2, 3, \dots$  and find the coefficients of

$$x, x^2, x^3, \dots \text{ and comparing with the given option as :}$$

$$\text{Coefficient of } x^2 \text{ is } = \frac{1}{3^3} + \frac{1}{3^2} + \frac{1}{3^1} = \frac{1}{3^3} \frac{[3^3-1]}{3-1} = \frac{13}{27};$$

$$\text{Which is given by option (a) } \frac{3^{n+1}-1}{2 \cdot (3^{n+1})} = \frac{3^3-1}{2 \cdot 3^3} = \frac{13}{27}.$$

5. (b)

$${}^{2n}C_n = \frac{2n!}{(n!)^2}$$

Coefficient will be

$$= \frac{1.3.5 \dots (2n-1) \cdot 2.4.6.8 \dots 2n}{(n!)^2} = \frac{1.3.5 \dots (2N-1)}{N!} \cdot 2^n$$

6. (b)

We have,

$$\{(1+x)(1+y)(x+y)\}^n$$

$$= (C_0 + C_1x + C_2x^2 + \dots + C_nx^n) \times (C_0y^n + C_1y^{n-1} + \dots + C_{n-1}y + C_n) \times (C_0x^n + C_1x^{n-1}y + \dots + C_{n-1}xy^{n-1} + C_ny^n)$$

Clearly, (Coefficient of  $x^n y^n$  on RHS)

$$C_0^3 + C_1^3 + C_2^3 + \dots + C_n^3 = \sum_{r=0}^n C_r^3$$

Hence (B) is correct answer.

7. (c)

$$\text{Put } x=y=z=1$$

$$\text{Sum of coefficient} = (1+2+1)^{10} = 4^{10}$$

8. (c)

$$\therefore \frac{T_2}{T_3} = \frac{2}{n-2+1} \cdot \frac{b}{a} = \frac{2}{n-1} \left(\frac{b}{a}\right) \text{ and}$$

$$\frac{T_3}{T_4} = \frac{3}{n+3-3+1} \cdot \left(\frac{b}{a}\right) = \frac{3}{n+1} \left(\frac{b}{a}\right)$$

$$\therefore \frac{T_2}{T_3} = \frac{T_3}{T_4} \text{ (given);}$$

$$\therefore \frac{2}{n-1} \left(\frac{b}{a}\right) = \frac{3}{n+1} \left(\frac{b}{a}\right) \Rightarrow 2n+2=3n-3 \Rightarrow n=5$$

9. (c)

Total number of terms in the expansion of  $(x+a)^{100}$  is 101, of which 50 terms get cancelled

10. (b)

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 \dots {}^nC_nx^n$$

on Integrating

$$\frac{(1+x)^{n+1}-1}{n+1} = \frac{{}^nC_1x^2}{2} + \frac{{}^nC_2x^3}{3} + \frac{{}^nC_3x^4}{4} \dots$$

Multiplying with  $x$  & differentiating

$$\frac{d}{dx} \left\{ x \left( \frac{(1+x)^{n+1}-1}{n+1} \right) \right\} =$$

$$\frac{d}{dx} \left\{ {}^nC_0x^2 + \frac{{}^nC_1x^3}{2} + \frac{{}^nC_2x^4}{3} + \frac{{}^nC_3x^5}{5} \dots \right\}$$

$$\frac{(1+x)^{n+1} + x(n+1)(1+x)^n - 1}{n+1} = \frac{{}^nC_1x^2}{2} + \frac{{}^nC_2x^3}{3} + \dots$$

.....

put  $x=1$

$$\frac{2^{n+1} + (n+1)2^n - 1}{n+1} = 2^n C_0 + \frac{3}{2} {}^nC_1 + \frac{4}{3} {}^nC_2 + \dots$$

$$= \frac{2^n(n+3)-1}{n+1}$$

11. (a)  $(5\sqrt{5} + 11)^{2n+1} = I + f$

Now suppose  $F = (5\sqrt{5} - 11)^{2n+1}$

$$I + f - F = (5\sqrt{5} + 11)^{2n+1} - (5\sqrt{5} - 11)^{2n+1}$$

$$= 2 [{}^{2n+1}C_1 (125)^n + {}^{2n+1}C_3 (125)^{n-1} 11^3 + \dots]$$

$$= 2 \text{ even integer}$$

$$\begin{aligned} I + f - F &= 2k \\ f - F &= 2k - I \text{ is an integer} \\ 0 \leq f < 1 & \quad 0 < F < 1 \\ -1 < -F < 0 \\ -1 < f - F < 1 \\ \Rightarrow f - F &= 0 \\ I = 2k &\Rightarrow \text{even integer} \end{aligned}$$

12. (b)

$$\begin{aligned} & {}^{47}C_4 + \sum_{r=1}^5 (52-r) C_3 \\ \Rightarrow & {}^{47}C_4 + {}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + {}^{48}C_3 + {}^{47}C_3 \\ & {}^{48}C_4 \\ \Rightarrow & {}^{48}C_4 + {}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + {}^{48}C_3 \\ & {}^{49}C_4 \end{aligned}$$

Similarly proceeding we get  ${}^{52}C_4$ .

13. (c)

$${}^5C_2 \left( \frac{1}{x} \right)^3 \left( x^{\log_2^x} \right)^2 = 10x^{2\log_2^x - 3} = 40$$

Middle term

$\Rightarrow$  Taking log of base 2 on both sides,

$$\text{let } \log_2^x = y$$

$$\Rightarrow (2y - 3)y = 2$$

$$\Rightarrow y = -1/2 \text{ or } 2$$

$$\Rightarrow x = \frac{1}{\sqrt{2}}, 4$$

14. (b)

$$\text{Sum of coeff} = (1 + 1)^n = 4096 \quad n = 12$$

$${}^{12}C_6 = \text{greatest Binomial coefficient} = 924$$

15. (c)

Put  $x = 1$  in equation .....(i)

$$0 = 1 + C_1 + C_2 + \dots + C_{12} \dots \dots \dots \text{(ii)}$$

Now put  $x = -1$  in equation (i);

$$2^6 = 1 - C_1 + C_2 - C_3 + \dots + C_{12} x^{12} \dots \dots \dots \text{(iii)}$$

Adding (i) & (ii)

$$2^6 = 2(1 + C_2 + C_4 + \dots + C_{12})$$

$$\therefore C_2 + C_4 + C_6 + \dots + C_{12} = 31$$

16. (d)

$$(3\sqrt{3} + 5)^7 = P + F$$

$$F' = (3\sqrt{3} - 5)^7$$

$$\therefore n = \text{odd. } F = F'$$

$$\begin{aligned} (P + F)F &= (3\sqrt{3} + 5)^7 (3\sqrt{3} - 5)^5 \\ &= (27 - 25)^7 \\ &= 2^7 \end{aligned}$$

17. (b)

We know that is  $r^{\text{th}}$   $(r + 1)^{\text{th}}$   $(r + 2)^{\text{th}}$  term coefficient are in an A.P. then  $m^2 - (4r + 1)m + 4r^2 - 2 = 0$  then for  $2^{\text{nd}}$   $3^{\text{rd}}$ ,  $4^{\text{th}}$  term  $r = 2$ ,  $m = 2n$   
 $(2n)^2 - 9 \times 2n + 14 = 0$   
 $2n^2 - 9n + 7 = 0$   
 $2n^2 - 9n + 8 = 1$

19. (b)

$$\begin{aligned} E &= (2n + 1)(2n + 3)(2n + 5) \dots (4n - 3)(4n - 1) \\ &= \frac{(2n)! (2n + 1)(2n + 2)(2n + 3)(2n + 4)(2n + 5) \dots (4n - 1)4n}{(2n)! (2n + 2)(2n + 4) \dots (4n)} \\ &= \frac{(4n)! n!}{(2n)! (2n)!} \end{aligned}$$

$$E = (2n)! n! 2^n (n + 1)(n + 2) \dots (2n)$$

$$E = \frac{(4n)! n!}{(2n)! (2n)!} \Rightarrow 2^n E = n! {}^{4n}C_{2n}$$

Hence  $2^n E$  is divisible by  ${}^{4n}C_{2n}$

20. (c)

$$\frac{1 - (1 - x)^n}{x} = \sum_{r=1}^n (-1)^{n-1} C_r x^{r-1}$$

Integrating between 0 and 1 yields the desired sum  $\sum_{r=1}^n \frac{1}{r}$